

A European public infrastructure for pharmaceutical research and development

The Covid-19 pandemic has highlighted some critical issues in how the pharmaceutical industry sets its research and development (R&D) priorities. Moreover, while the European pharmaceutical industry sector is strong in economic terms, it does not benefit in R&D terms from a comparable public infrastructure to the United States federal government-sponsored National Institutes of Health (NIH) or the Biomedical Advanced Research and Development Authority (BARDA). This gap exacerbates some structural issues in the European context, as European Union (EU) institutions have limited leverage to address market failures affecting the functioning and regulation of the EU pharmaceutical market.

In a nutshell, the failures for which the current public policies and regulatory remedies are less than adequate, are: 1) Disconnection between corporate R&D choices and public health priorities; 2) Mismatch between largely publicly funded open-science practices in fundamental research and the patent system protecting investors; 3) Excess returns for financial investors in the pharmaceutical industry arising from government-subsidised R&D; 4) Oligopolistic market power on the supply side, and issues of access and affordability of medicines; 5) Inadequate medicines optimisation studies following market authorisation; 6) Information asymmetries in the public procurement of medicines.

This situation suggests a need to explore a new policy approach based on a more direct public intervention, as was successfully experienced in space policy and other science-based sectors: the creation of a new European player with a public mission in pharmaceutical R&D and biomedical innovation; and a long-term vision and funding, called European medicines infrastructure (EMI).

Four policy options are put forward here to set up the EMI. They arise from the combination of two dimensions: the scope of the mission (narrow versus wide) of the EMI on one side, and its internal R&D capacity (limited versus extended) on the other.

Option 1: A mission-oriented, European R&D infrastructure focused on infectious diseases and acting as a public procurement hub for biomedical innovation

The EMI would work to its own agenda, specifically in the highest priority field of R&D on vaccines and medicines for infectious/transmissible disease and arrangements for their delivery. The new organisation should have its own governance (with top-level scientific and managerial skills), budget, and a core, but relatively limited, of in-house R&D laboratories. It would essentially work through R&D contracts with selected third parties. Such contracts would not be designed as grants or subsidies to third parties, but rather as public procurement arrangements, with the intellectual ownership rights of any discoveries and the delivery mechanisms of new medicines under the ultimate responsibility of the new EMI.

This option aims to ensure a coordinated EU approach to addressing the market and policy failures identified in the field of infectious disease. The rationale is to promote a coordinated agenda of large projects in areas affected by private sector under-investment, but where problems could be tackled by acting in concert and mobilising a critical mass of funds. The new organisation should identify the

range of R&D projects in infectious disease, in agreement with stakeholders such as the European Medicines Agency, the European Centre for Disease Prevention and Control (ECDC), the European Health Emergency Preparedness and Response Authority (HERA), the European Research Council (ERC), the European Innovation Council (EIC), or the European Molecular Biology Laboratory (EMBL), and other biomedical research bodies, to avoid overlap. Implementing such an agenda and delivering new medicines would be the core responsibility of the new organisation, which could act through procurement contracts with academia, existing R&D institutions, and pharmaceutical companies.

Unlike many existing European initiatives, this option would not involve creating an executive agency or a funding body that allocates funds through competitive calls to many small-size projects with a loose connection to its wider research agenda. Instead, it involves creating a planning, management and delivery organisation that defines its own long-term portfolio of R&D projects and enters contracts with third parties to implement them. In this context, it is worth noting that unlike HERA, the EMI (in all its configurations) would aim at structural change in the public interest of the current model of pharmaceutical R&D.

According to the European Commission communication (COM(2021) 576 final), HERA will act as an enabler of strategic R&D projects on vaccines and medicines for infectious disease by pooling capacities and creating a long-term and large-scale EU platform for multi-centre clinical trials. However, HERA apparently will not have the responsibility, resources and capacities to directly implement a strategic pharmaceutical R&D projects pipeline in the public interest.

Managing a portfolio of innovative R&D projects over 30 years is precisely the main mission of the EMI (in all its configurations). HERA would need to rely mainly on the current pharmaceutical players. As a result, it would not gain the critical mass to structurally shift pharmaceutical companies' and other players' R&D choices towards public-health priorities, apart from a limited intervention area related to possible emergencies.

In the absence of the EMI, the prioritisation and allocation of EU funds for pharmaceutical R&D would continue to follow a 'grant'-based approach, which risks dispersing funds towards a myriad of relatively small projects, and the capture of EU funds by existing organisations and pharmaceutical companies, each with their own agendas, without clear added value. The proposed EMI may be seen as complementary to HERA, with shared public health values and different missions and tools.

Indicatively, the annual budget of the EMI under Option 1 could be set equal to that of the NIH Intramural Research Program, which amounts to about €3.5 billion. Given this annual budget, considering overheads and capital cost, and taking the R&D cost per drug of about €1 billion per project as a benchmark, the EMI may be expected to deliver a total of 80-100 biomedical innovations from 2023 to 2050 under Option 1, including vaccines, medicines and biotechnologies. The budget could further increase if non-EU countries joined the EMI, for example with mission-specific additional budgets supported by different coalitions of governments.

Option 2: A mission-oriented, European R&D infrastructure with a diversified R&D portfolio beyond infectious diseases, acting as a public procurement hub for a broad range of biomedical innovations in all areas inadequately covered by the industry

The second option is similar to the first, but with a wider mission. Under this option, the scope of the EMI would include other fields suffering both public and private sector under-investment, such as vaccines and medicines for infectious disease, but also medicines related to neurodegenerative conditions, some types of cancer and genetic conditions, rare diseases, and health issues in low-income countries. It would work around missions designed by 'horizontal' R&D concepts, technologies and platforms.

As in the previous option, the new organisation should have its own governance (with both scientific and managerial skills), budget, contractual arrangements with external suppliers and partners, and a core but relatively limited in-house laboratory and staff capacity. It would mainly work with a range of public procurement contracts for innovation with third parties around the identified horizontal missions. Indicatively, and as for Option 1, the annual budget and the outcome of the EMI under Option 2 could be set at €3.5 billion and 80-100 innovative medicines/technologies, respectively.

Option 3: A large-scale, mission-oriented, European R&D infrastructure focused on infectious diseases that runs projects within its labs, collaborating flexibly with third parties

The third option concerns the creation of a large-scale, mission-oriented, EMI with an exclusive focus on infectious disease, but, unlike the previous two options, this new organisation, while also working through contracts with third parties, would employ its own scientific staff working in world-class, dedicated laboratories to primarily manage its research in-house. In the rich EU context of excellent research centres, a hub-and-spoke model would be most appropriate for this new infrastructure. The model consists of a central hub (that may be identified in existing infrastructure), to which decentralised but integrated laboratories are connected. The EMBL constitutes a reference in this regard.

While performing in-house research, the new organisation would be largely open to R&D vaccine and therapy collaborations for diseases arising from viruses and other pathogens, including research on pathogens resistant to existing antibiotics. This work could take place in partnership with third-party research centres at the national or European level and with selected pharmaceutical companies (those that are seriously willing to invest in this area), even outside the EU when needed.

Such collaborations would be based on clear and transparent contractual arrangements, including on securing the intellectual property of discoveries for the EMI in the public interest and on marketing authorisation. A significant advantage of this option is that the EMI owns (either wholly or in specific cases within public-private partnerships) the results of the R&D projects it carries out and manages its intellectual property rights exclusively in the public interest.

Indicatively, the annual budget of the EMI under Option 3 could be equal to the budget of the European Space Agency for 2021, amounting to nearly €6.5 billion (including contributions to specific missions from some participants). Given this annual budget, considering overheads and capital cost, and taking the R&D cost per drug of about €1 billion per project as a benchmark, the EMI could be expected to deliver a total of 130-150 innovative medicines/technologies from 2023 to 2050 under Option 3. As for Options 1 and 2, the suggested annual budget could further increase if non-EU countries joined the EMI, for example with mission-specific additional budgets supported by different coalitions of governments, as per the current functioning of ESA.

Option 4: A large-scale, mission-oriented, European R&D infrastructure with a diversified R&D portfolio that runs projects within its labs and with flexible collaborations with third parties

The fourth option is the most ambitious in terms of scope and delivery mechanisms. It is similar to Option 3, as it concerns the creation of a large-scale, mission-oriented European R&D infrastructure that manages R&D projects mainly in-house within its own laboratories. It would, however, have a wider R&D agenda (like Option 2), i.e. would not be constrained to research on infectious disease compared to the previous option.

The history of discovery in medicines, vaccines and other biotech innovations (including for diagnostics and materials) suggests that science-based advances in different fields spring from new ideas and technologies with unexpected outcomes and scope. A notable example is mRNA vaccines for Covid-19,

built on scientific and technological advances in molecular biology initially understood as supporting a broad range of new therapeutic approaches involving the production of certain proteins.

In this perspective, therapies for cancer or neurodegenerative diseases are not completely different from therapies for or vaccines against infectious disease. This is just one example of why the scientific community would prefer a more flexible R&D mission for the proposed infrastructure, as envisaged in Options 2 and 4. Indicatively, and as for Option 3, the annual budget of the EMI under Option 4 could be equal to €6.5 billion. Accordingly, the total outcome in 30 years could be about 130-150 innovative medicines/technologies.

If supported by a long-term strategic commitment on the part of the European Commission and the Member States, this option would create the biggest public R&D infrastructure in the world. Indeed, under Option 4, the EMI would have a scale greater than the US federal government-sponsored NIH's Intramural Research Program and would go beyond it in terms of ownership and the delivery mechanisms for innovative medicines and related technologies.

It would place Europe firmly as the top global player in the field of R&D for medicines, with direct benefits for patients, public health systems and early career researchers, and also with potential benefits for the European pharmaceutical industry, especially small and medium-sized enterprises, in terms of possible partnerships on specific projects.

This document is based on the STOA study 'European pharmaceutical research and development: Could a public infrastructure overcome market failures?'. The study was written by Massimo Florio, University of Milan, with Chiara Pancotti (CSIL Centre for Industrial Studies), at the request of the Panel for the Future of Science and Technology (STOA). STOA administrator responsible: Gianluca Quaglio. The authors of this briefing are Massimo Florio, University of Milan, Italy, and Chiara Pancotti, Centre for Industrial Studies, Milan, Italy.

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